Pipeline Congestion and Basis Differentials

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Abstract In the U.S., natural gas pipeline transport has undergone a wave of deregulatory actions over the past several decades. The underlying motive has been the presumption that removing regulatory frictions would facilitate spot price arbitrage, helping to integrate prices across geographic locations and improve efficiency. Yet certain frictions, specifically the effect of congestion on transportation costs, inhibit positive deregulatory impacts on efficiency. With the increase in domestic production and consumption of natural gas over the coming decades, upward pressure on the demand for transport will likely result in an increased occurrence of persistently congested pipeline routes. In this paper we explore the relationship between congestion and spot prices using a simple network model, paying particular attention to the influence of storage. We find that as congestion between two hubs increases, the scarcity value of transmission capacity rises, driving a wedge between spot prices. We empirically quantify this effect over a specific pipeline route in the Rocky Mountain region that closely resembles our structural design. Although our results paint a stark picture of the impact that congestion can have on efficiency, we also find evidence that the availability of storage mitigates the price effects of congestion through the intertemporal substitution of transmission services.

Keywords natural gas pipelines; congestion; storage; spot prices; secondary markets

JEL Codes Q41; Q48; R41

Revised, April 2014

1 Introduction

Starting in the late 1970s, deregulatory actions have sought to better facilitate interstate commerce in the U.S. natural gas market. A primary rationale has been that freer markets would accommodate arbitrage opportunities, thereby integrating spot prices across regions.[1] However, pipeline congestion is not uncommon, and can undermine market efficiency by way of greater transportation costs. This, in turn, adversely impacts opportunities for spot price arbitrage.[2] An interesting complication is that this increase in costs does not originate from actions taken by the pipeline companies themselves. The Federal Energy Regulatory Commission (FERC) limits the primary market price of capacity, preventing pipeline companies from realizing higher returns from competing bidders (Marmer et al. 2007). There are two consequences. First, there is a potential for infrastructure constraints (i.e. bottlenecks) to emerge or persist.[3] Second, any scarcity rents that result from congestion are captured by primary purchasers of pipeline capacity via *unregulated secondary markets*: i.e. spot price-based gas transactions or releases of unused capacity at an unregulated rate. These frictions work counter to an efficient market mechanism.

Network congestion is costly, and capacity constraints can magnify congestion problems. Nearly all transportation networks (highways, railroads, gas and electric utilities, etc.) are subject to capacity constraints over specific routes. The idea that congestion on a capacity-constrained network increases transportation costs is not limited to the natural gas pipeline network. DeVany

[1] Several researchers found evidence of convergence in regional gas spot prices prior to 2000 (De Vany and Walls 1993, 1994a,b; Walls 1995; Serletis 1997; and Dahl and Matson 1998), arguing that local, regional, and national gas markets evolved in response to increased arbitrage opportunities. Finnoff et al. (2004) find further evidence that FERC Order No. 636 spurred changes in pipelines' operational and financial behavior that reduced 'balkanization', increased competition, and reduced expense preference behavior.

[2] Where delivery constraints between major trading hubs exist, prices at the trading hubs can be impacted. This effect occurs irrespective of distance. De Vany and Walls (1995, p. 46) state, "…if there is no link [between markets] or if there are limits on the flow of the commodity over the link, then the prices of the commodity can move farther away from each other, especially in short time periods."

[3] Vickrey (1969) defines a bottleneck as "a situation in which a network segment has a fixed capacity substantially smaller relative to flow demand than that of preceding and succeeding segments."

and Walls (1999) investigate spot price co-integration across 11 electricity markets in the Western United States, noting that line losses and congestion imply transportation cost over a given arc on the network is an increasing and convex function of flow. Insufficient capacity relative to transport demand also generates scarcity rents for those with rights to that transport capacity, resulting in wealth transfers from commodity producers and consumers to the owners of scarce capacity. Atkinson and Kerkvliet (1986) found empirical evidence that railroads captured a significant proportion of the potential rents on low-sulfur Wyoming coal, and that rents shifted to the railroads following their deregulation in 1980. These inter-related effects of constrained capacity, congestion costs, and rent extraction limit the ability of spot price arbitrage to integrate prices across geographically distant locations.

As interstate pipeline capacity is a key factor in determining the amount of natural gas that can be physically traded between markets, the relationships between capacities, flows, and spot prices are both systematic and measurable. Over the long-run, the trade-off between greater capacity and higher congestion costs is similar to the 'adequacy problem' in the market for electrical generation capacity.[4] In the short-run, however, a pipeline operating at capacity is unable to satisfy short-run transport demand in excess of its capacity limit, leading to potentially severe market distortions.

In this paper we model and quantify the impact of pipeline capacity constraints upon natural gas spot price arbitrage. In the U.S. increased reserves, advances in extraction technology, and expanding consumption (Energy Information Administration [EIA] 2010a) have put upward pressure on the demand for pipeline transport. As a result, bottlenecks on certain segments of the pipeline network can constrain deliveries, thereby driving a wedge between the

[4] For example, see Cramton et al. (2013) for a complete discussion of the adequacy problem as pertains to electrical generation capacity, and Bowring (2013) for a more focused analysis of adequacy in the PJM electrical generation capacity market.

prices at trading hubs on either side of the constraint. Following MacAvoy (2007), we call the difference in spot prices of gas at two trading hubs the 'basis differential'. To examine the determinants of these differentials, we adapt the hypotheses of existing network models to include the effects of congestion and the influence of storage. We then test the predictions using a unique dataset of price and pipeline data from the Rocky Mountain region. Our results show that as the pipeline route between two hubs becomes congested, the basis differential between their spot prices widens, sometimes dramatically so.[5]

There are broad implications associated with the persistence of wide basis differentials given the volume of natural gas transactions affected by spot prices. In 2009, of the nearly 56 trillion cubic feet (Tcf) of natural gas physically transacted in the U.S., approximately 22% of that volume, over 12 Tcf, was transacted at daily index prices (FERC 2010). Our empirical estimates show that over a single transport route in the Rocky Mountain region, a mere 3.5 percent increase in flow inflates the basis differential between two local hubs' spot prices by nearly 23 percent, implying an estimated increase in average monthly transport costs between the two hubs of roughly $315,000.

The remainder of the paper proceeds as follows. Section 2 provides an overview of the market(s) for interstate natural gas pipeline capacity and transport, including discussion of the current federal regulatory structure. Section 3 describes the theoretical foundation for our analysis and the associated testable hypotheses. The empirics are presented in Section 4, along

[5] Importantly, the effect of congested transport infrastructure on price differentials is not limited to the natural gas pipeline network. A similar and widely publicized example had been occurring between two key oil price indices in the U.S. Starting in early 2011, increased oil production in Canada and the central U.S. overwhelmed the pipeline infrastructure transporting oil from the Cushing, OK hub to the Gulf Coast. This bottleneck resulted in a price differential between the West Texas Intermediate (Cushing) and Brent Crude (Gulf Coast) indices that averaged roughly $20 per barrel, and persisted through mid-2013. However, once new pipeline and rail links between Cushing and the Gulf Coast came online, the differential shrunk to a manageable $6 per barrel on average (DiColo 2013).

with further discussion of the results and their implications for the natural gas market and pipeline regulation. Section 5 concludes and offers possibilities for continuing research.

2 The Primary and Secondary Markets

The two tiers in the market for natural gas pipeline transport capacity follow recent FERC regulations. FERC Order No. 636 requires *open access* transportation on natural gas pipelines,[6] which are regulated according to a rate-of-return framework. In the primary market, entities wishing to have *guaranteed* access to transport capacity (most prominently local distribution companies [LDCs] and gas marketers) purchase 'firm' capacity contracts at FERC-regulated rates. As primary market sales of firm capacity occur prior to the construction of the pipeline, firm customers can in some sense be thought of as investors in the infrastructure asset. Before a new pipeline can be constructed, FERC requires the pipeline firm to demonstrate in its application that long-term (10 years or longer) firm capacity contracts are in place as evidence of market necessity and to underwrite the financing of the project (Interstate Natural Gas Association of America [INGAA] 2009; Black and Veatch LLC 2012). Once the pipeline is in operation, an unregulated secondary market then allows owners of firm contracts to recover their capacity's underlying market value over time. They may either utilize capacity to transact gas, or release capacity rights to other shippers.[7]

[6] Two early empirical studies (Hollas 1994; 1999) examined the impacts of FERC's push toward open access pipeline transport and restructuring of the natural gas market on public utility pricing. Following implementation of FERC Order No. 636 (as well as its pre-cursor, Order No. 436), industrial customers enjoyed significant reductions in retail gas rates relative to residential and commercial users.

[7] Prior to the passage of Order No. 636, Alger and Toman (1990) presented experimental evidence that a market-based approach for this class of transaction could "outperform traditional rate-setting regulation" used in interstate pipeline transmission, with the caveat that short-term resale rates could greatly exceed the regulated primary market rates during peak demand periods.

At the secondary market's inception, FERC required released capacity to be priced at the regulated primary market rate. This restriction gave capacity holders little incentive to release it during peak demand periods. However, agents quickly discovered they could circumvent FERC's rules on capacity release by employing privately negotiated "buy-sell" arrangements, in which firm capacity holders buy available supply from producers in order to sell to downstream buyers.[8] In 2008 FERC approved Order No. 712, explicitly relaxing all restrictions on pricing in the formal capacity release market (FERC 2008; INGAA 2009; McGrew 2009, p. 123). This change transformed the secondary market into a competitive spot market for capacity. Formal capacity release transactions are reported via FERC-mandated websites, allowing secondary market participants to observe transaction values. LDCs and other regulated entities are required to release capacity via FERC's formal capacity release system. Unregulated gas marketers, however, have more flexibility to engage in legal buy-sell transactions than regulated utilities[9] and have become significant players in the secondary market.

At any point in time, the entire capacity of a pipeline is reserved via firm contracts (FERC 1999). Because pipeline capacity is fixed in the short run, agents wishing to purchase firm contracts in the primary market for capacity over a given route find themselves unable to do so. Open-access implies that anyone can utilize a pipeline to ship gas. However, because the pipeline's entire capacity is under contract, any would-be shippers not owning firm capacity

[8] See Tussing and Tippee (1995, p. 231) for a complete discussion. Certain types of buy-sell transactions are prohibited by FERC. For example, a capacity holder cannot buy from a seller with intent to resell to a pre-specified buyer after transport. This is considered a violation of open access policy. FERC's 'shipper-must-have-title' rule requires a shipper to own any gas transported on the pipeline (FERC 2012). Firm capacity owners wishing to exploit a constraint must either release unused capacity directly to shippers, or buy the gas commodity from suppliers, ship, and then resell to any willing buyers at the destination market price.

[9] Personal communication (April 20, 2012) with Gregory Lander, President of Skipping Stone, LLC energy consulting group.

must acquire transportation services via the secondary market.[10] The (per-unit) payment made by shippers to firm contract holders for the utilization of capacity is effectively a charge for transportation. It is unregulated, allowing firm capacity holders to exploit the scarcity of capacity available in the secondary market. Intuitively, shippers bid up these charges as unused transmission capacity becomes scarce. Because the primary market two-part tariff is regulated, firm contract holders are able to extract scarcity rents whenever the transportation charge exceeds the primary market two-part tariff. However, the opposite also applies—when transmission capacity available in the secondary market is plentiful, the transportation charge may fall short of the primary market two-part tariff, introducing the risk that the cost of the firm capacity contract is not fully recovered. Potential primary market participants will vary in their levels of risk aversion, and a larger primary market commitment to capacity exposes a contract holder to greater risk, limiting the amount of capacity to which any given agent would care to commit. The risk acts as a constraint on primary market demand and as a sorting mechanism between the primary and secondary markets. Highly risk-averse agents are less likely to purchase primary market contracts, preferring to participate as buyers of transportation services in the secondary market. Conversely, more risk-tolerant agents are willing to purchase long-term primary market contracts in order to utilize or release their capacity in the secondary market.[11] The exogenous distribution of risk-aversion levels across market participants, along with the short-run nature of our analysis, allows us to define the number of primary market participants (and the amount of capacity they are willing to contract) as exogenous.

[10] Shippers do have direct access to the pipeline via 'interruptible transport', the rate for which is also regulated by FERC. However, this service is by nature less reliable, as transmission may be interrupted at any moment by a firm claim on capacity (McGrew, 2009).

[11] For large industrial users, for whom transmission contract costs can be passed on directly to final consumers, a higher degree of risk-aversion might lead to a greater willingness to contract firm capacity for reasons of reliability of supply. But for unregulated gas traders, who by nature face greater risk from fluctuations in both demand conditions and uncertain contract cost recovery, there are significant primary market risks.

The intuitive foundation for our analysis is that secondary market scarcity is particularly acute for routes characterized by insufficient capacity relative to transportation demand, leading to significant opportunities for rent extraction. Because ours is a static short-run model of the network equilibrium any point in time, the pipeline's capacity is fixed and assumed to be fully reserved in the primary market by an exogenously determined number of contract holders.[12] As spot transactions are by nature made within a short-run time horizon, transportation of spot transacted gas is thus more prone to being confined to whatever capacity is available in the secondary market. For this reason we model the basis differential as being equivalent to the per-unit transportation charge that limits the ability of arbitrage to fully align spot prices across the network.

3 Pipeline Model

We follow Cremer and Laffont (2002) and Cremer, Gasmi, and Laffont (2003) in modeling a simple pipeline network as shown in Figure 1. There are three segments in the network. Segment 'w' runs from Production Basin 1 to End-user Market w (one can think of 'w' here as referring to 'west'). Segment 'e' runs from Production Basin 2 to End-user Market e (one can think of 'e' here as referring to 'east'). Segment 'b' runs between the two production basins, and maintains a strictly eastbound gas flow. Storage is available at Hub 1.[13] The geographical relation between the key players suggests a vertical structure, in which gas sellers at Hub 1 deliver gas to buyers at Hub 2. An alternative approach is to envision a model of the buyer-seller

[12] Additionally, we assume the regulated primary market rates to be fixed over the time period under consideration, and have verified this to be the case in our empirical application.

[13] We augment the Cremer, Gasmi, and Laffont (2003) network model by allowing for storage, as resource firms have incentive to hold inventories to smooth production over time when prices are stochastic and sufficiently volatile (Mason 2010). In practice, storage plays a vital role in facilitating the use of natural gas through hedging and network balancing (INGAA 2009). That point noted, our central focus is on the manner in which pipeline transportation costs impact the markets at Hubs 1 and 2; the simplified model we discuss below is able to produce the testable hypotheses of interest.

interactions as taking place at Hub 2, and being subject to transactions costs that influence the ability of sellers to bring the good in question to market. This simpler interpretation allows us to develop the key hypotheses we test later in the paper, and so we will adopt this model in the pursuant discussion.

In our interpretation of the network model, the transactions costs associated with using the pipeline are akin to an excise tax upon (upstream) sellers. In the pipeline network model of Cremer, Gasmi, and Laffont (2003), when there is a competitive secondary market for pipeline capacity, the spot price at a hub is the sum of the production (supply) price, the transportation charge, and the (secondary) spot price of capacity. We include storage as an additional factor, because it tempers fluctuations in the spot price of capacity, and thus mediates spot price fluctuations at the hubs.[14] Transportation charges and the cost of capacity, net of the influence of storage, are transactions costs that drive a wedge between the price paid by buyers at Hub 2 and the net price received by sellers at Hub 1. Any exogenous event that increases the transportation cost or the spot price of capacity will increase this wedge, further raising the downstream price and lowering the upstream price net of the transactions costs.

Changes in these determinants of transactions costs are related to the volume of gas flowing through the pipeline in segment b. We assume the pipeline's capacity is fully reserved in the primary market via firm contracts, and that producers in each production basin sell extracted gas through their adjacent hubs.[15] Consumers in each end-user market purchase the

[14] This mediation by storage was demonstrated in the peak-load literature (Nguyen, 1976; Gravelle, 1976; Crew and Kleindorfer, 1979). The ability to store reduces the price differential between peak and off-peak demand periods. Hollas (1990) found empirical support for this effect in the natural gas pipeline transmission industry, using firm and interruptible LDC transmission rates as proxies for peak and off-peak prices.

[15] We assume that production in each basin is exogenous, reflecting the conventional wisdom that natural gas production is generally price inelastic in the short-run (International Energy Agency [IEA] 1998, p. 36; Krichene 2002). The idea is that wells that are actively producing are operated at production capacity, and that the costs associated with shutting in production from a natural gas well, and then reopening the well later, are typically too great to justify doing so in response to small variations in price.

gas commodity through their adjacent hubs. Demand conditions in each end-user market, indicated by their market prices, are exogenous. Capacity owners mediate gas transactions, resulting in aggregate flows on each segment of the pipeline. As the volume flowing through the pipeline increases, the opportunity cost of remaining space also rises, which induces the increase in transactions costs. Storage allows owners of gas to avoid some of these increases in transactions costs: by storing gas they are able to wait for more favorable conditions.

In this setting, the secondary market allocates scarce transmission capacity. For simplicity, we do not distinguish between the formal capacity release market and the 'buy-sell' market values of transmission capacity, because both are strongly related to the prevailing spot price basis differential. Our goal is to analyze the influence of pipeline congestion and storage on the basis differential. We assume that the number of firm contract holders is large enough to render the secondary market competitive, in line with the FERC's assessment that there is sufficient competition in this market (FERC 2009).

At any point in time t, each segment of the pipeline $i = w, b, e$ is subject to a capacity constraint on flows, namely that the volume of gas flowing through the segment, $y_{i,t}$, cannot exceed the capacity $K_{i,t}$.[16] Cremer, Gasmi, and Laffont (2003) analytically demonstrate that the spot price of capacity is a function of the shadow value of the capacity constraint. Our interpretation is that this relation is similar to a derived inverse demand function; in equilibrium it is linked to market conditions across the network. As flows approach the capacity constraint, the spot price of capacity rises. Arbitrage implies that the basis differential between the spot prices of natural gas at Hubs 1 and 2 is equal to the spot price of the capacity linking them. It

[16] In practice, a pipeline's 'capacity' is defined as the maximum throughput per unit time (typically expressed in daily increments) that can be maintained over an extended interval, and is subject to various technological, safety, and regulatory constraints. Capacity does *not* refer to the maximum physical throughput capability of the system or segment, which can greatly exceed the pipeline's certificated capacity (www.eia.gov).

follows that the basis differential increases as congestion increases and pushes the pipeline to capacity, or (holding flows constant) is reduced by a capacity expansion.[17] We thus define the basis differential on segment b at time $t, \tau_{b,t}$, as a function of capacity and flows: $\tau_{b,t}(K_{b,t}, y_{b,t})$, where $\partial\tau_{b,t}/\partial K_{b,t} < 0$ and $\partial\tau_{b,t}/\partial y_{b,t} > 0$. Storage allows owners of gas to avoid shipping when available transmission capacity is scarce and the temporary cost of congestion is highest. Stored gas can instead be withdrawn and sent along the bottleneck segment once congestion eases and capacity costs decline.

4 Empirical Analysis

Other researchers have studied natural gas price behavior to infer whether a pipeline bottleneck exists between two markets (Brown and Yücel 2008; Marmer et al. 2007). In contrast, we are interested in i) the magnitude of a bottleneck's influence over prices, and ii) the magnitude to which the availability of storage dampens this influence. In particular, we estimate the effects on spot prices of congestion at a known bottleneck in a regional setting where storage is available. We assume that each observation of an endogenous variable in our data set is representative of an equilibrium at a particular point in time. Estimation of a multi-equation system allows us to observe how endogenous variables respond in equilibrium to each other and to key exogenous variables.

The Rocky Mountain regional pipeline network provides an excellent template for our empirical study. The Opal Hub in southwest Wyoming and the Cheyenne Hub along the Colorado-Wyoming border are approximately 325 miles apart and are connected by three pipelines whose combined capacity is currently about 3.2 million MMBtu/day. Any volume of

[17] While spot traded volumes are presumably decreasing in the spot price at the source hub and increasing in the spot price at the destination hub or end-user market, overall traded volumes need not respond in kind, particularly when spot traded volumes are small in proportion to overall traded volumes.

gas transacted between these hubs must be shipped through these pipelines, which are operated by Colorado Interstate Gas (CIG), Wyoming Interstate Company (WIC), and Rockies Express (REX). The two hubs are as far upstream as possible: while Cheyenne lies downstream of Opal there is no other hub upstream of Opal. Accordingly, they represent the headwaters of the Rocky Mountain natural gas supply system. There is a storage facility near the Opal Hub (Clay Basin, which is located along the Wyoming-Utah border). With a total capacity of roughly 117.5 Bcf (51.2 Bcf in working inventories), it is among the ten largest natural gas storage facilities in the U.S. (FERC 2013). Figure 2 shows the Rocky Mountain regional network.

The Rocky Mountain region produces a significant amount of the U.S. supply of natural gas. At over 5 Tcf of production, it was the nation's top producing region in 2009 (EIA 2011). Wyoming's total proved reserves in that year were estimated at 36.75 Tcf (EIA 2010b). All production from the state's largest producing basin, the Upper Green River Basin (which contains the Jonah and Pinedale fields), is sent directly to the Opal Hub. A smaller source, the Powder River Basin (PRB), sends most of its production directly to the Cheyenne Hub. Stable pipeline flows enable us to pinpoint constraint locations and clearly identify the bottleneck. The flow of gas between Opal and Cheyenne is strictly eastbound, with Clay Basin lying to the west (i.e. upstream) of the bottleneck. Six pipelines fan out eastward from Cheyenne.[18] All other spokes not connecting the two hubs flow either strictly west (from Opal) to the Pacific Coast or strictly east (from Cheyenne) to the Midwest.

[18] We should note that two major regional export pipelines begun operations in 2011. Opening of these lines lies outside our sample period, and they are not shown in Figure 2.

4.1 Description of the Data

Natural gas pipeline companies are required by FERC to maintain an "electronic bulletin board" on which they continually update real-time data about capacities and gas flows for all segments/locations on their systems. Pipeline data were gathered for daily operational capacities and scheduled flow volumes for several important pipeline locations.[19] These correspond to export points 1 through 10 and the three bottleneck pipeline segments shown in Figure 2.[20] We use the *IntraDay2* gas cycle, representing a pipeline's most updated and accurate posting. All observations are converted to consistent units, MMBtu/day. Table 1 lists the ten export points, as well as the bottleneck locations on the three connecting pipelines (each of which lies between Opal and Cheyenne).

Daily spot price series were purchased from Platts (McGraw-Hill). For all spot prices, we use the daily index midpoint.[21] Regional production, storage, and consumption data were purchased from Bentek Energy, LLC.[22] Our sample period ranges from May 8, 2007 to October 29, 2010. After removing all non-trading days (weekends and holidays), we have 1,119 observations. We separate our sample into two cohorts: days when the Cheyenne price exceeds the Opal price, which we term "Cohort 1," and days when the Opal price exceeds the Cheyenne price, which we term "Cohort 2."[23] Cohort 1 makes up the majority of our sample, with 80.6%

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[19] Capacities and scheduled flows are defined in units of volume per period. For natural gas, the generally applicable time period is one day, and volume is either posted in 1000's of cubic feet (Mcf), or is converted into millions of British thermal units (MMBtu). The conversion ratio is roughly 1.02/1 for Mcf/MMBtu. Although rare, on occasion scheduled volume can exceed maximum certificated capacity.

[20] Export points 11 and 12 in Figure 2 are considered external to our system, and are thus omitted from the empirical analysis. The bottleneck pipeline segments are not numbered in Figure 2, but their collective location is indicated.

[21] The daily index midpoint is simply the midpoint between the high and low recorded spot prices on a given day. This is the value that is typically reported by industry newsletters (for example, Platts *Gas Daily*).

[22] Due to copyright law, the terms and conditions of the purchase agreements prohibit publication and/or sharing of purchased data. Please contact the corresponding author for authentication.

[23] Intuitively, this separation is related to the likelihood that differing technological and institutional factors govern upstream versus downstream sales. Our simple model relies on the assumption that the upstream price must exceed the downstream price for the transportation charge over a given segment to be positive. We thus consider days when the Cheyenne price exceeds the Opal price to be representative of our model's design. In reality, this is not

of the observations, while Cohort 2 contains 17.4% of the observations.[24] Table 2 presents summary statistics for the Opal and Cheyenne spot prices and their basis differential—first for the pooled dataset, and then for each cohort separately.

Figure 3 demonstrates the congestion-basis differential relationship for each cohort. Observations from Cohort 1 are illustrated by dots, and observations from Cohort 2 are illustrated by diamonds. The vertical axis measures the mean basis differential,[25] and the horizontal axis contains different ranges of unused capacity at the bottleneck.[26] Listed next to each marker is the number of observations used to calculate the mean. For example, the dot in the top-left corner indicates that there are 187 observations in Cohort 1 where unused capacity at the bottleneck is less than 50,000 MMBtu. The mean basis differential over this sub-sample is remarkably high—roughly 62 cents. The rents available to firm capacity owners during these periods of high congestion were large in relation to regulated tariffs: during the sample period CIG, REX, and WIC had regulated two-part tariffs totaling \$0.337, \$0.235, and \$0.098 per MMBtu per day, respectively. We observe a strong negative relationship between the mean basis differential and unused capacity at the bottleneck in Cohort 1. We regard this as convincing evidence of the relationship between the transportation charge and congestion. In contrast, there is no clear relationship between unused capacity at the bottleneck and the basis differential for observations in Cohort 2.

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always the case, although the Opal price exceeding the Cheyenne price certainly seems to be the exception rather than the rule.

[24] The 22 days on which these prices are equal are considered uninformative and are ignored in our formal empirical analysis. This is due to the fact that they do not fit into either of the other two cohorts, and because the small sample size makes testing the subset on its own difficult with respect to formal empirical inference.

[25] Calculated as the Cheyenne spot price minus the Opal spot price (presented using absolute values in Figure 3).

[26] Measured as the difference between daily operating capacity and scheduled flow volume, which we consider to be a reasonable measure of congestion over the bottleneck route.

4.2 Model Specification and Estimation Procedure

We estimate a seven-equation system. The equation for endogenous variable y_j ($j = 1, \ldots, 7$) at time *t* is

$$y_{j,t} = Y_{j,t}\beta + X_{j,t}\gamma + Y_{j,t-s}\delta^s + Z_{j,t}\varphi + \epsilon_{j,t}, \tag{1}$$

where $Y_{j,t}$ is a vector of other endogenous variables, $X_{j,t}$ are exogenous explanatory variables, $Y_{j,t-s}$ is a vector containing s lags of $y_{j,t}$, $Z_{j,t}$ are dummy variables described below, and $\epsilon_{j,t}$ is a random error term. β, γ, δ^s, and φ are coefficient vectors.

The seven endogenous variables are the natural logs of the spot prices at Opal (p_t^o) and Cheyenne (p_t^c),[27] the basis differential (τ_t),[28] aggregate scheduled volumes $\left(y_{i,t}, i = w, b, e\right)$, and the net change in storage at Clay Basin ($stor_t$).[29] The regressors for each equation are chosen based on geographical location. We assume that an endogenous variable is directly affected only by those variables that are geographically adjacent. Indirect effects of geographically non-adjacent variables are assumed to occur via the appropriate adjacent variables' simultaneous direct effects.

The empirical model includes a number of exogenous variables. First, our analysis indicates that capacity plays an important role. Because decisions to expand capacity at a point in time are not impacted by any price at that moment, but by expected future revenues, capacities can be treated as exogenous. Accordingly, we regard operating capacity over each of the three segments of the network ($K_{i,t}, i = w, b, e$) as an exogenous variable. We also include two

[27] Using natural logs can be thought of as allowing for non-linear effects in demand curves. While there are other published price indices in the region, including Northwest Wyoming Pool, Northwest South of Green River, and White River, they are all are considered to fluctuate closely with the Opal Hub price.
[28] Calculated as the net of spot prices $p_t^c - p_t^o$, in levels, implying that the Cohort 2 basis differential is recorded as negative in our data.
[29] There is no storage facility near Cheyenne. CIG has a significant amount of system storage, however it all lies to the east and southeast of Denver. For this reason, we do not consider this system storage to be connected to the Cheyenne Hub.

western and two eastern price points that we consider to be exogenous to our system. The Los Angeles (Kern River, delivered) city gate price (p_t^{la}) represents an anchor demand price for the major market in Southern California. An important competing supply for westbound gas coming from Opal is Kingsgate Center (p_t^{kg}), servicing Canadian imports into the Pacific Northwest. The Chicago city gate price (p_t^{chi}) represents demand conditions in the nearest eastern metropolitan market for which a price index is available. Henry Hub (p_t^{hen}) is widely considered to be the primary supply hub in the nation. Statewide consumption for Utah (ut_t), Wyoming (wy_t), and Colorado (co_t) are included to control for local demand.[30] Aggregate production west of the bottleneck, comprised of the Overthrust, Greater Green River, and Piceance basins, is given by q_t^w. Production from the PRB, which is transported directly to the Cheyenne Hub and which lies to the east of the bottleneck, is q_t^e. Production arising from the Big Horn and Wind River Basins in central Wyoming, and which enters the system to the west of the bottleneck, is q_t^c.

To control for time-series effects that might influence our estimates, each equation contains four lags of its left-hand side variable.[31] Potential structural breaks that may have coincided with major capacity expansions at the bottleneck are tested using time dummies (z_t^1, z_t^2), for (i) dates prior to the 686.3 MMcf/day expansion that occurred January 2, 2008, and (ii) dates between the January 2, 2008 expansion and the 262.7 MMcf/day expansion on June 1,

[30] We consider statewide demand for Colorado as a geographically eastern variable only. Roughly 85% of Colorado's population lives in the eastern half of the state (www.colorado.gov), and all gas coming from the west is routed through the Cheyenne Hub.

[31] To assess the appropriate lag structure, we estimated models containing zero, one, two, three, four and five lags. For each of these structures, we obtained the Akaike Information Criterion and Bayesian Information Criterion from the underlying 2SLS estimation. Based on these statistics, we inferred that the optimal lag structure contained four lags for each cohort. In an evaluation of 2SLS and 3SLS estimators with structural dynamic models of non-stationary and 'possibly' cointegrated variables, with unknown unit roots or rank of cointegration, Hsaio and Wang (2007) show that the 2SLS t-statistics and 3SLS z-scores of individual coefficients are asymptotically distributed as standard normal random variables.

2009. Accordingly, the baseline corresponds to dates after June 1, 2009. We also include seasonal dummies, with $f = 1$ for observations in fall (0 otherwise), $w = 1$ for observations in winter (0 otherwise), and $spr = 1$ for observations in spring (0 otherwise).

Our estimation procedure starts by applying two-stage least squares (2SLS) to our empirical model. Using these results, we test for first- and second-stage under-identification, first- and second-stage weak identification, and over-identification; we also include other second-stage endogeneity and exogeneity tests.[32] Various identification test statistics from this step are summarized in the Appendix: Table A.1 for Cohort 1 and Table A.2 for Cohort 2. These statistics broadly support the hypothesis that our identification strategy is valid.[33] Having confirmed that our model is properly structured, we then obtain final parameter estimates using three-stage least squares (3SLS).[34]

4.3 3SLS Estimation Results

Estimation results for the system of equations in (1) based on observations from Cohort 1 are reported in Table 3. For ease of location in Tables 3 and 4, in what follows we identify equation-by-equation coefficients in the text according to left-hand side (LHS) and right-hand side (RHS) variables. The Table 3 estimates indicate that reductions in congestion at the bottleneck, which can result from either an increase in capacity (RHS: K) or a decrease in scheduled flows (RHS: y^b), drive the Opal price (LHS: p^o) up and the Cheyenne price (LHS: p^c) down, decreasing the

[32] The equation for flows on the bottleneck pipeline segment, y_{bt}, is used as the central equation for our 2SLS instrumental variables procedure, as it contains all other endogenous variables as regressors.

[33] While the first- and second-stage weak identification tests are inconclusive, each of the three weak-instrument robust inference tests pass at 99% confidence; we therefore conclude that weak identification is not a major cause for concern in our estimation.

[34] Due to the simultaneous nature of our system of equations, 3SLS is the natural estimation procedure for obtaining coefficient estimates. One caveat with using 3SLS is that coefficient estimates may be inconsistent if there is any serial correlation in the error structure. Although the errors may display some contemporaneous correlation across equations, we have no reason to suspect any other correlations or heteroskedasticities in the error structure, and thus maintain confidence in the validity of our 3SLS procedure.

basis differential (LHS: τ) between them. These results provide compelling evidence that pipeline congestion affects spot prices via the transportation charge levied upon shippers. We also find that additions to storage exert a negative effect on the Cohort 1 basis differential (LHS: τ, RHS: $stor$), which is loosely consistent with the peak-load literature, although this effect is not quite statistically significant.

Storage, in conjunction with the potential bottleneck constraint, plays an important role. The peak-load literature has already shown that storage downstream of a bottleneck alleviates the price effects of congestion, as it allows downstream demand to be met despite transmission constraints. Our analysis indicates that upstream storage has value as well, as it allows shippers to avoid high transportation costs through the intertemporal substitution of transmission services. On days when the bottleneck constraint is tight, gas can be stored; on such days, the Opal price will be lower, the Cheyenne price higher (shown in the storage column of Table 3). When the constraint is relaxed, gas that had been stored can be delivered through the bottleneck, thereby avoiding the higher tariffs implicitly resulting from congestion costs. That storage additions exert a negative effect upon the basis differential for Cohort 1 is consistent with this view.[35] The reciprocal effect, corresponding to the negative coefficient on the *basis differential* in the storage equation (LHS: $stor$, RHS: τ), is indicative of spot price arbitrage—as the Cheyenne price increases relative to the Opal price, some gas is withdrawn from storage (provided the bottleneck is not operating at full capacity) in order to take advantage of the arbitrage opportunity.[36]

[35] That this reduction is not statistically significant may be an artifact of the feature that the bottleneck constraint does not bind every day. On days when there is no congestion, other motivations for storage will trump any incentives to intertemporally arbitrage, weakening the significance of the impacts from days with tight constraints.

[36] The remaining determinants of storage at Clay Basin (LHS: $stor$) conform to logic. Storage increases as production in the western basins increases (RHS: q^w). It is drawn down (i) in the winter (RHS: w); (ii) when outflows west and through the bottleneck increase (RHS: y^w, y^b); and (iii) as consumption in Utah increases (RHS: ut).

Regression results based on observations from Cohort 2 are reported in Table 4. These estimates indicate that capacity at the bottleneck exerts a statistically significant effect on the basis differential (LHS: τ, RHS: K), whereas flows do not (RHS: y^b). These observations suggest that an entirely different process governs the system on during days in Cohort 2, which we discuss in greater detail below. For this cohort, additions to storage (RHS: $stor$) do not exert a statistically significant effect on prices (LHS: p^o, p^c) at the two hubs separately. However, we find that storage has a negative influence on the Cohort 2 basis differential (LHS: τ, RHS: $stor$). Because the Cohort 2 basis differential is a negative number, this implies, perhaps counter-intuitively, that additions to storage are associated with wider Cohort 2 basis differentials.

The explanation for this result relates to the simple operational fact that gas flows continuously from West to East. In Cohort 2 the bottleneck exerts no influence because upstream sales are not subject to the same physical constraints as are downstream sales. Any shipper wishing to arbitrage the Cohort 2 basis differential by trading gas from Cheyenne upstream to Opal will transact that shipment via a reduction in the net eastbound flow through the bottleneck.[37] Such shipments would not be affected by any binding capacity constraint at the bottleneck, nor would they be influenced by the associated scarcity value of available transmission capacity. They would, however, be influenced by storage, because deliveries to storage from sellers at Cheyenne imply greater demand for upstream transmission. As additions to storage increase, the Cohort 2 basis differential widens, as shown by the estimates in the basis differential equation. As the Cheyenne price rises, this stimulates releases from storage, which in turn act to lower the price of upstream transmission. As in Cohort 1, that the use of storage is motivated in Cohort 2 by a desire to intertemporally arbitrage prices is evidenced by the negative

[37] In other words, one would arrange for additional purchases from other sources at Cheyenne, for example the Powder River Basin, and reduce injections into the pipeline at Opal, thereby freeing up the extra gas to be sold at Opal.

influence of the Opal price and positive influence of the Cheyenne price upon additions to storage (though these effects are statistically insignificant): as the Opal (Cheyenne) price rises (falls), stored gas is sent through the bottleneck for sale.

Our estimates suggest that spot transactions do have some effect on overall flows. The positive coefficient on the basis differential in the Cohort 1 bottleneck flow equation (Table 3, LHS: y^b, RHS: τ) is indicative of spot price arbitrage—as the basis differential widens between the two hubs, flows through the bottleneck increase. For Cohort 2, this coefficient is again positive (implying that flows through the bottleneck decrease as the Cohort 2 basis differential widens), but not statistically significant (Table 4, LHS: y^b, RHS: τ). In contrast, the relationships between individual spot prices and flows can be explained by demand and supply shocks. For example, a positive correlation in Cohort 1 between outflows from Opal and its spot price (Table 3, LHS: p^o, RHS: y^w, y^b) is likely due to demand shocks. When demand for gas from Opal is high, both prices and outflows are high. Conversely, a negative correlation between flows through the bottleneck and the Cheyenne price (Table 3, LHS: p^c, RHS: y^b) is indicative of a negative supply shock in the form of less gas being delivered from Opal. The price at Cheyenne is higher when less gas is supplied via the bottleneck. Predictably, for Cohort 2 these effects are reversed for flows through the bottleneck (Table 4, LHS: p^c, RHS: y^b).

A key implication of our analysis is the 'basis stabilization' effect of additional pipeline capacity. Holding flows constant, an increase in maximum capacity at the bottleneck reduces the basis differential in both cohorts. This is plainly observable in the data by plotting the Cheyenne-Opal basis differential time series (Figure 4). The two major capacity expansions corresponding to dummies z^1 and z^2 are marked by solid vertical lines. An intermediate expansion of 68.6 MMcf/day occurred on January 14, 2009, and is marked by a dashed line. The basis stabilization effect is most noticeable following the first and largest expansion, but dissipated somewhat as flow demand increased.[38] Notice also the relative stability of prices after June 1, 2009. Table 5 summarizes the data underlying the basis stabilization effect. Because sufficient capacity relative to flow demand has been in place between these two hubs for the past few years, arbitrage has no longer been inhibited by congestion pricing, and we do not observe such large and persistent divergence in spot prices in the last part of our sample.

Holding maximum capacity constant, an increase in flow demand would increase the basis differential. We illustrate the magnitude of the effect through a simple thought experiment. Holding all variables at their Cohort 1 means over the time interval following the second major capacity expansion, our estimation accurately predicts the sub-sample mean basis differential of roughly 7 cents (see Table 5).[39] Average unused capacity for this sub-sample is slightly over 400,000 MMBtu/day. Relative to maximum capacity, our estimates indicate that an increase in flow demand at the bottleneck of 100,000 MMBtu/day, or roughly a 3.5 percent increase, would sufficiently increase congestion as to increase the mean basis differential by 1.6 cents (Table 3, LHS: τ, RHS: y_b). Assuming that 22 percent of all physical transactions are made using spot prices (FERC, 2010), this increase in the mean basis differential implies a monthly increase of $316,129 in average transport costs over this bottleneck route alone.[40]

This illustration of the potential increase in costs associated with greater congestion is particularly relevant given the likely steady increases in natural gas production and consumption

[38] It is also important to note that prices were at their peak in 2008, adding to the propensity for high basis differentials relative to other intervals in our sample.

[39] Applying our Cohort 1 coefficient estimates in Table 3 to the system of equations (2), and setting all variables equal to their Cohort 1 means following the second major capacity expansion, we obtain a basis differential of 7 cents.

[40] Average scheduled volume for the sub-sample is 2,837,330 MMBtu/day. Using the FERC estimate, 646,213 MMBtu/per day of that volume would be transacted using spot prices. Multiplying this value by 1.6 cents, the increase in the basis differential we estimated in the text, yields an estimated increase in total transport costs of $10,399 per day. Multiplying that by the average number of days in a month (30.4) equals $316,129.

over the next two decades. The EIA (2010a) projects an increase in total annual production from roughly 22 Tcf in 2009 to almost 27.5 Tcf in 2035. Consumption is expected to grow by 0.6 percent per year over this period (EIA 2010a; EIA 2011), largely due to an anticipated increase in electricity production using natural gas. Our estimates of the effect of congestion on transport costs over one pipeline suggest important potential welfare effects. Extrapolating to the national level, potential increases in congestion over multiple routes in the natural gas pipeline network seem likely to have substantial negative aggregate welfare impacts. Furthermore, the apparent ability of storage to mitigate the price effects of congestion through intertemporal substitution of transmission services suggests that increases in storage capacity will also be crucial for maintaining market efficiency as transportation demand on the national pipeline network continues to rise.

A subtler question concerns whether the diversion of congestion rents from the pipeline to owners of firm capacity distorts the economic signal for the pipeline network to install additional capacity when and where needed. This possibility appears to be at odds with the observation that there were two major expansions in the pipeline network between Opal and Cheyenne during our sample period. What economic incentives led to these expansions? Gilbert and Harris (1984) model oligopolistic competition over the scale and timing of 'lumpy' investment in new capital when the production technology exhibits increasing returns to scale—a scenario descriptive of natural gas pipeline transmission capacity. A key result of the Gilbert-Harris model is that with non-trivial lag times between firms' investment decisions and the construction of new plant, the firm with the shortest lag time has an advantage. Specifically, while that firm has an incentive to expand due to rising demand, it can increase rents on existing productive capacity by delaying construction.

The imposition of a fixed rate-of-return (ROR) on the pipeline's profits would appear to render this result irrelevant—under ROR regulation the pipeline would not receive a strong market signal to expand and would be indifferent between build times. But in the pipeline scenario, the real investors in expanded capacity are the purchasers of the associated firm capacity contracts. Thus, the incentive leading to capacity expansion arises from the scarcity rents that are available to those seeking to invest in additional firm capacity, rather than to the pipeline company itself. The potential for these incentives to motivate pipeline expansion is blunted by the regulatory regime governing pipeline tariffs. The Gilbert-Harris result applies to entrants to the firm capacity market, because lag times are presumably long due to extensive permitting requirements, and because rents on firm capacity are not constrained. In a loose oligopoly of firm contract owners with no constraints on secondary market rate setting, high demand on the part of pipeline users would deliver the appropriate expansion incentives. However, a distortion occurs when firm contract owners cannot respond to positive expected profits by offering higher tariffs to the pipeline because tariffs are capped under ROR regulation. This observation does not imply that pipeline capacity would be expanded more rapidly if ROR regulations were lifted and the rents accrued to the pipeline instead of the firm capacity owners. It does imply that the pipeline owners are unlikely to receive the economic signal to expand pipeline capacity. Instead, the purchasers of firm capacity receive the signal, and (in the Gilbert-Harris sense) have incentive to invest in expanded capacity due to growing demand, but also to delay that investment in order to increase rents.

Figure 5 provides evidence supporting the perspective just described. Here, we plot time series of maximum capacity and scheduled volume at the bottleneck before and after the first (and largest) capacity expansion. Clearly, the route was persistently congested for a full nine months leading up to the expansion: the average capacity utilization rate over that period was 97.5%; on 103 out of 190 days the capacity utilization rate was at 99% or above; and on 78 of those 190 days capacity was 100% utilized. This effect was even more dramatic in the four months prior to the expansion, as the utilization rate was at 100% on 64 of these 82 trading days. Clearly, transportation volumes over this particular route were truncated for several months by capacity constraints. Referring back to Figure 4, basis differentials were exceptionally high, providing significant rents to firm capacity owners.

The underlying intuition is that, with ROR regulation in the primary market and an unregulated secondary market, market power (and the expansion decision) is transferred from the pipeline to the firm capacity purchasers. Thus, regulated tariffs in the primary market prevent monopoly (or oligopoly) pricing, but distort expansion signals. These distorted signals are a cost of preventing the exercise of market power by the pipeline; if it seems unlikely that tariffs would be marked up significantly because of market power, then the distorted signals seem likely to have the more important welfare effect.

5 Conclusion

In this article we have explored the relationship between pipeline congestion and natural gas basis differentials. Owners of firm transport capacity operate in an unregulated secondary market, completing transactions in which the scarcity value of pipeline transport is either explicitly known (in the case of the formal capacity release market), or is tacitly built into the commodity pricing agreement (in the case of legal buy-sell transactions). Using data based on a two-hub network in the Rocky Mountain region, we have shown how congestion affects spot prices through a deregulated secondary market in which primary owners of firm capacity contracts capture the true market value of scarce transmission capacity. As congestion between the hubs increases, the transportation charge for shipping services rises, driving a wedge between spot prices. This conclusion confirms the intuition in Cremer, Gasmi, and Laffont (2003), that the spot price of transmission capacity should be increasing as congestion increases. Additionally, we have demonstrated that storage provides a valuable service even when upstream of the bottleneck constraint—extending the results established in the peak-load literature. Specifically, upstream storage allows shippers to avoid high transportation charges via intertermporal substitution of transmission services. This result suggests that storage capacity, whether upstream or downstream of transmission constraints, will play a crucial role in maintaining market efficiency as pipeline transmission demand increases over the coming decades.

The importance of natural gas in the U.S. domestic energy portfolio has grown dramatically, which points to a concern that the national pipeline network may increasingly experience constrained flows. This, in turn, implies potentially significant benefits from expanding capacity to alleviate such constraints. It has been estimated that investments totaling from $160 to $210 billion will be needed over the next 20 years to finance additions averaging 1,200 to 1,300 miles per year (INGAA 2009). If investment in pipelines is unable to keep pace with the growing market, the likelihood is high that more transportation routes will exhibit the bottleneck phenomenon, in which case the availability of storage will be a primary determinant of the degree to which the congestion effects of such bottlenecks interfere with spot price integration.

Our analysis seems likely to have broader relevance, beyond the natural gas industry. For example, with the rise of globalization international shipping is becoming increasingly important. As any spatially distributed market with geographically separated supply and demand centers must be served by a physical transport system, the capacity of specific routes and storability of the transported commodity seem likely to have measurable effects on market prices.

One can envision several extensions to this research. First, although we feel our model adequately describes system behavior for the smaller subset of our data for which the daily spot price at Opal exceeds that at Cheyenne, further investigation is needed to identify alternative impediments to transacting gas upstream that might affect the basis differential on these days. Second, as roughly half of all gas transactions are based on monthly price indices (FERC 2010), it is worth investigating whether the congestion-price relationship also holds for monthly price indices. Third, our short-run analysis suggests a need for a long-run model of capacity expansion. Important issues to be resolved include the optimal scale and timing of capacity expansion. Fourth, one could analyze a multiple-hub system, allowing for feedback loops throughout the network. A key question here is: how does regional export capacity affect local prices, relative to national prices?

Finally, and perhaps most importantly, we hope to understand how the current regulatory environment—price controls in the primary market for capacity juxtaposed against a fully deregulated secondary market—affects both short-run and long-run behavior. Current research is underway that supports the notion that the rate-of-return regulatory framework imposed by FERC may indeed reduce investment in capacity relative to what is socially optimal. If the regulatory environment does stifle investment in capacity, then the congestion effects on prices outlined in this paper would be exacerbated as a result.

Acknowledgements This article has greatly benefited from the help and insights of David Aadland, Brian Jeffries, Erik Johnson, Gregory Lander, Jason Shogren, Alexandre Skiba, Brian Towler, and Aaron Wood. Two anonymous referees provided useful input, and pressed us to clarify our arguments and contribution. Earlier versions were presented at the 13th Annual CU Environmental and Resource Economics Workshop, Vail, CO (Oct. 7-8, 2011), the 2012 Occasional Workshop in Environmental and Resource Economics, Santa Barbara, CA (Feb. 24-25, 2012), and the 2012 Association of Environmental and Resource Economists (AERE) Summer Conference, Asheville, NC (June 4-5, 2012). We thank the participants of those events for helpful comments and observations. The School of Energy Resources at the University of Wyoming provided financial support for this research.

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Tables & Figures

Table 1 Selected Interstate Pipeline Locations

Map Location (Figure 2)	Company Name	Location/Segment
Westbound Out of Opal		
1	Northwest Pipeline	Kemmerer Compressor
2	Kern River Interstate Gas Co.	Veyo Compressor
3	Northwest Pipeline	La Plata B Compressor
4	TransColorado Pipeline	LOC Segment 220
Eastbound Out of Cheyenne		
5	Colorado Interstate Gas	Kit Carson Compressor
6	Southern Star Central Gas Pipeline	St. Francis Compressor
7	Cheyenne Plains Interstate Gas Co.	Cheyenne Plains East
8	Kinder-Morgan Interstate Gas Transmission	LOC Segment 190
9	Trailblazer Pipeline Co.	LOC Segment 10
10	Rockies Express	LOC Segment 200
Bottleneck Locations		
B1	Colorado Interstate Gas	Laramie East
B2	Rockies Express	LOC Segment 150
B3	Wyoming Interstate Co.	Laramie East

Table 2 Summary statistics for Opal and Cheyenne spot prices

	Pooled	**Cohort 1 (Cheyenne > Opal)**	**Cohort 2 (Opal > Cheyenne)**
Observations	**1119**	**902**	**195**
Mean Opal price ($)	**4.543**	**4.586**	**4.355**
St. Dev.	*1.906*	*1.815*	*2.25*
Mean Cheyenne price ($)	**4.735**	**4.845**	**4.256**
St. Dev.	*1.954*	*1.857*	*2.273*
Mean Basis Differential ($)	**0.192**	**0.259**	**0.099**
St. Dev.	*0.412*	*0.427*	*0.138*
Basis Differential Distribution			
10^{th} percentile	-0.035	0.02	0.01
25^{th} percentile	0.015	0.045	0.02
50^{th} percentile	0.065	0.10	0.045
75^{th} percentile	0.205	0.27	0.12
90^{th} percentile	0.57	0.755	0.26
95^{th} percentile	1.04	1.19	0.359

Basis differentials calculated as Cheyenne (midpoint) minus Opal (midpoint).
Cohort basis differentials listed in absolute values.

Table 3 3SLS Estimation Results (Cohort 1)

RHS Var.	Equation (LHS Variable)						
	p^o	p^c	τ	y_w	y_b	y_e	$stor$
p^o		0.5949*** (22.41)	-1.2344*** (-10.46)	72,290.95 *** (4.23)	486,814.5*** (4.81)		-979,473.2*** (-2.65)
p^c	0.8869*** (16.23)		1.3407*** (10.46)		-463,942.6*** (-4.23)	-32,605.01 (-1.50)	1,117,139** (2.48)
τ					99,605.5*** (3.28)		-218,597.2** (2.24)
y_w	-3.93×10^{-9} (-0.10)				-0.1047** (-2.31)		-0.3864*** (-2.79)
y_b	-5.12×10^{-8} (-1.16)	5.12×10^{-8} * (1.85)	1.61×10^{-7} *** (4.46)	-0.1324*** (-3.87)		0.3245*** (8.77)	-0.4619*** (-3.67)
y_e		-3.01×10^{-8} (-1.54)			0.0342 (1.45)		
$stor$	-1.14×10^{-8} (-0.32)	-1.58×10^{-8} (-1.08)	-3.45×10^{-8} (-1.56)	0.1730*** (4.95)	0.1291*** (3.31)		
K_w	8.33×10^{-8} ** (2.08)			0.1579*** (4.70)			
K_b	1.03×10^{-7} *** (5.11)	-1.01×10^{-7} *** (-5.44)	-1.98×10^{-7} *** (-6.13)		0.0663** (2.25)		
K_e		5.64×10^{-8} *** (4.03)				-0.0151 (-0.29)	
p^{la}	0.0875 (1.62)			-22,565.95 (-0.44)			-188,630.4 (-1.09)
p^{kg}	-0.1361*** (-2.80)			-48,940.57 (-0.96)			127,008.1 (0.78)
p^{chi}		0.3032*** (4.71)				-90,663.93 (-0.82)	
p^{hen}		-0.1292** (-2.16)				118,383.9 (1.12)	
q^w	-2.06×10^{-8} (-0.54)			0.1649*** (5.35)	0.2134*** (7.45)		0.4876*** (3.92)
q^c		-4.40×10^{-8} (-0.45)			0.1864 (1.34)	-0.0624 (-0.37)	
q^e		-7.31×10^{-8} *** (-2.67)				0.4186*** (9.64)	

Table 3 (cont'd)

RHS Var.	Equation (LHS Variable)						
	p^o	p^c	τ	y_w	y_b	y_e	$stor$
ut	-2.24×10^{-8}			0.1298***	0.0689		-1.1227***
	(-0.48)			(3.26)	(1.52)		(-11.04)
wy	-2.32×10^{-7}	-4.03×10^{-8}		0.0776	-0.2820	-0.6665***	-0.7046
	(-1.00)	(-0.25)		(0.39)	(-1.24)	(-2.59)	(-1.01)
co		2.75×10^{-9}				-0.1440***	
		(0.28)				(-10.18)	
f				-20,899.89*	-13,642.44	-14,341.99	-26,818.46
				(-1.73)	(-1.12)	(-1.05)	(-0.62)
w				40,864.57**	63,386.58***	-36,421.61**	-233,550.3***
				(2.49)	(3.42)	(-2.17)	(-4.13)
spr				-31,372.25***	-8,804.80	-10,361.59	-4,639.95
				(-3.33)	(-0.86)	(-0.90)	(-0.14)
z^1				99,103.25***	-5,106.55	-117,087.3	
				(4.37)	(-0.11)	(-1.27)	
z^2				52,621.16***	-15,309.22	1,478.32	
				(4.31)	(-0.87)	(0.07)	
lag 1	0.1830***	0.1043***	0.2541***	0.6175***	0.5426***	0.6160***	-0.1482***
	(6.46)	(5.26)	(12.03)	(16.79)	(15.65)	(20.50)	(-4.77)
lag 2	0.0932***	-0.0011	0.0161	0.0038	0.0478	0.0470	-0.0495*
	(3.76)	(-0.06)	(0.83)	(0.10)	(1.47)	(1.40)	(-1.70)
lag 3	-0.0206	0.0340*	0.0759***	0.0025	0.0354	-0.0005	-0.0177
	(-0.94)	(1.65)	(3.61)	(0.07)	(1.05)	(-0.01)	(-0.62)
lag 4	-0.0769***	0.0668***	0.0771***	0.0999***	0.0487*	-0.0010	0.0411
	(-3.65)	(3.60)	(4.13)	(3.06)	(1.76)	(-0.03)	(1.46)
$const$	-0.3518**	0.1615***	0.0646	-308,959.0**	-357,097.8***	388,072.2	388,072.2
	(-2.40)	(5.15)	(1.26)	(-2.20)	(-2.94)	(1.24)	(1.24)
Equation Statistics[a]							
R^2	0.92	0.95	0.77	0.66	0.97	0.98	0.49
χ^2	9,995.74	19,457.22	2,225.05	2,633.87	33,857.50	40,796.61	925.81
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Prices in logs. z-statistics in parentheses below coefficient estimates. [a] p-values in parentheses below χ^2 statistics.

* Indicates 90% significance. ** Indicates 95% significance. *** Indicates 99% significance.

Table 4 3SLS Estimation Results (Cohort 2)

RHS Var.	Equation (LHS Variable)						
	p^o	p^c	τ	y_w	y_b	y_e	$stor$
p^o		1.2970*** (39.69)	-0.1388 (-1.53)	62,506.10*** (3.70)	-185,284.7** (-2.23)		-458,812.1 (-1.21)
p^c	0.7919*** (43.88)		0.1331* (1.69)		217,723.6*** (3.14)	32,806.07** (2.20)	409,423.1 (1.35)
τ					149,159.7 (1.29)		-106,471.8 (-0.18)
y_w	-1.41×10^{-8} (-0.48)				-0.2531*** (-3.28)		-0.8769*** (-2.60)
y_b	-6.00×10^{-8} (-1.54)	1.09×10^{-7} ** (2.22)	1.20×10^{-8} (0.25)	-0.2140*** (-4.00)		0.3410*** (6.14)	-0.8402*** (-2.80)
y_e		-9.25×10^{-9} (-0.40)			0.0038 (0.08)		
$stor$	-1.86×10^{-8} (-0.79)	1.93×10^{-8} (0.68)	-4.50×10^{-8} * (-1.72)	-0.0767** (-2.55)	0.0560* (1.70)		
K_w	-7.47×10^{-9} (-0.34)			0.0428 (0.80)			
K_b	4.38×10^{-8} (1.52)	-7.17×10^{-8} * (-1.64)	7.14×10^{-8} * (1.68)		0.0272 (0.46)		
K_e		-7.93×10^{-9} (-0.36)				-0.0421 (-0.77)	
p^{la}	0.0254 (0.93)			48,323.99 (0.62)			249,542.4 (0.70)
p^{kg}	0.0956*** (3.04)			-51,378.69 (-0.65)			-83,460.91 (-0.24)
p^{chi}		-0.2245*** (-2.65)				-220,526.6 (-1.32)	
p^{hen}		0.0621 (0.79)				187,665.3 (1.12)	
q^w	3.56×10^{-9} (0.16)			0.2849*** (6.57)	0.0327 (0.55)		0.9147*** (3.22)
q^c		1.38×10^{-8} (0.09)			0.0778 (0.25)	0.4986 (1.48)	
q^e		-1.53×10^{-8} (-0.26)				0.2295* (1.88)	

Table 4 (cont'd)

RHS Var.	Equation (LHS Variable)						
	p^o	p^c	τ	y_w	y_b	y_e	$stor$
ut	-1.48×10^{-8}			-0.2281***	-0.0077		-1.5314***
	(-0.57)			(-4.00)	(-0.12)		(-5.26)
wy	-2.10×10^{-7}	2.23×10^{-7}		0.8064**	-0.2850	-1.8227***	0.3349
	(-0.76)	(0.58)		(2.07)	(-0.66)	(-4.59)	(0.20)
co		7.88×10^{-9}				-0.0840***	
		(0.15)				(-3.03)	
f				-33,417.54*	13,790.16	33,963.93*	-118,595.8
				(-1.77)	(0.69)	(1.69)	(-1.35)
w				29,522.68	2,146.16	-13,618.49	-124,645.2
				(1.17)	(0.08)	(-0.54)	(-1.03)
spr				19,307.22	-15,190.17	-34,559.96*	-32,581.76
				(1.03)	(-0.76)	(-1.75)	(-0.36)
z^1				117,658.4***	31,302.25	-168,991.6*	
				(2.69)	(0.31)	(-1.73)	
z^2				31,585.26	38,041.20	-8,702.82	
				(1.43)	(1.20)	(-0.28)	
lag 1	0.0586**	-0.0836***	0.0743*	0.5365***	0.5869***	0.4096***	-0.1727**
	(2.51)	(-2.74)	(1.69)	(9.08)	(8.53)	(6.61)	(-2.28)
lag 2	0.0729**	-0.0622***	-0.0088	-0.0055	0.0412	0.1999***	-0.0361
	(2.51)	(-3.98)	(-0.19)	(-0.08)	(0.57)	(3.21)	(-0.42)
lag 3	-0.0766***	0.0364**	-0.0430	-0.1732**	0.0327	-0.0125	-0.0855
	(-3.76)	(2.56)	(-1.30)	(-2.41)	(0.55)	(-0.25)	(-1.00)
lag 4	-0.0025	0.0261**	0.0252	0.1922***	0.0499	0.0844**	0.0770
	(-0.20)	(2.27)	(0.87)	(3.50)	(0.94)	(2.20)	(0.92)
$const$	0.1387	-0.0550	-0.2890***	242,698.7	120,943.8	367,380.0	719,046.5
	(1.31)	(-0.69)	(-5.80)	(0.90)	(0.51)	(1.59)	(0.87)
Equation Statistics[a]							
R^2	0.98	0.98	0.31	0.73	0.97	0.99	0.46
χ^2	11,640.25	9,168.85	78.33	542.75	8,305.93	15,759.34	167.38
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Prices in logs. z-statistics in parentheses below coefficient estimates. [a] p-values in parentheses below χ^2 statistics.

* Indicates 90% significance. ** Indicates 95% significance. *** Indicates 99% significance.

Table 5 Spot price behavior before and after major capacity expansions

	Before Jan. 2, 2008	Jan. 2, 2008 to Jun. 1, 2009	After Jun. 1, 2009
Cheyenne spot price			
Cohort 1 mean	5.10	5.62	3.73
Cohort 2 mean	3.20	5.76	3.46
Opal spot price			
Cohort 1 mean	4.63	5.43	3.66
Cohort 2 mean	3.38	5.84	3.49
Basis Differential			
Cohort 1 mean	0.47	0.19	0.07
Cohort 2 mean	0.18	0.08	0.03

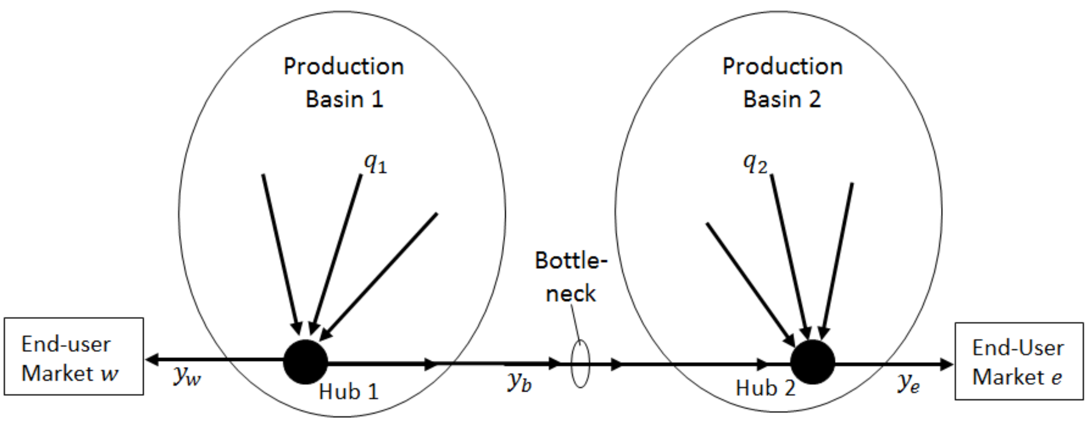

Fig. 1 Two-hub, one-pipeline system with distinct production basins and end-user markets

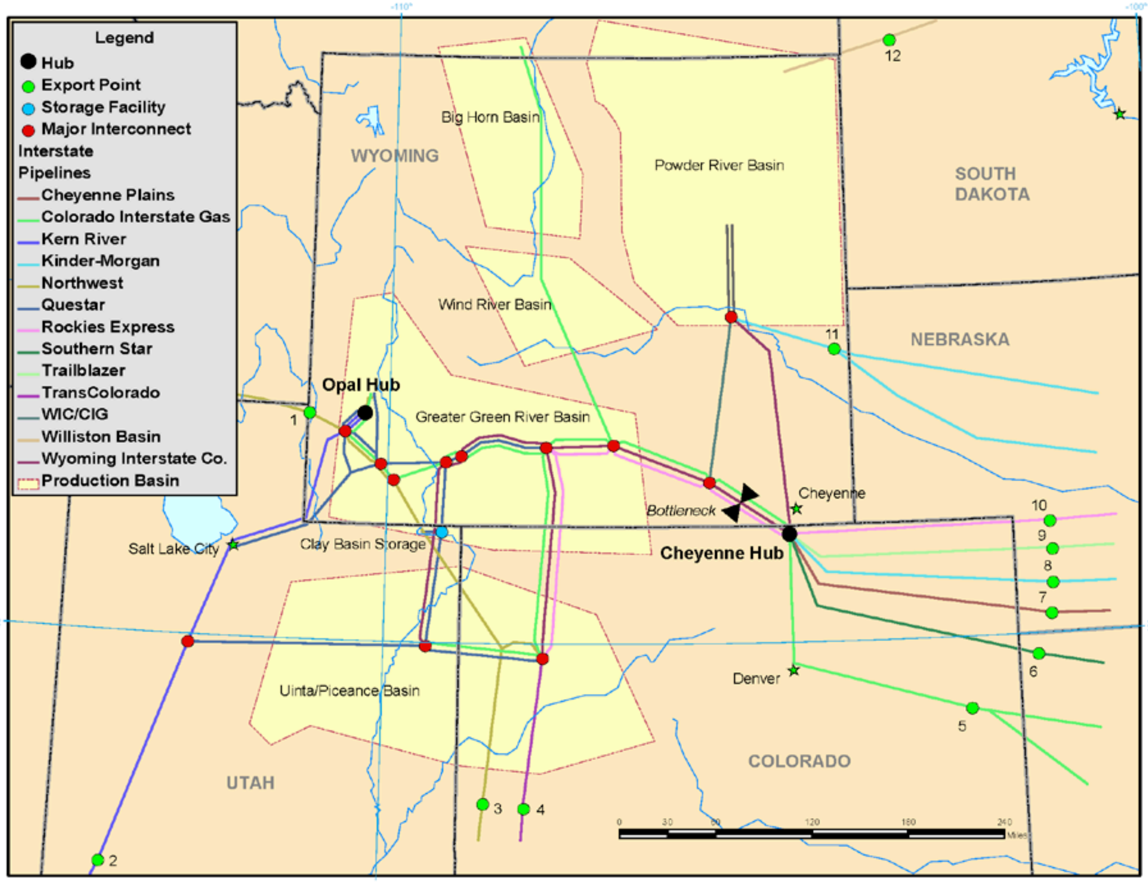

Fig. 2 Rocky Mountain interstate pipeline network (geographical locations are approximate). Sources: CIG System Map (2007), EIA, Questar System Map (2011), Questar-Overthrust System Map (2011), WIC System Map (2011), Wyoming Oil and Gas Conservation Commission, Wyoming Pipeline Authority

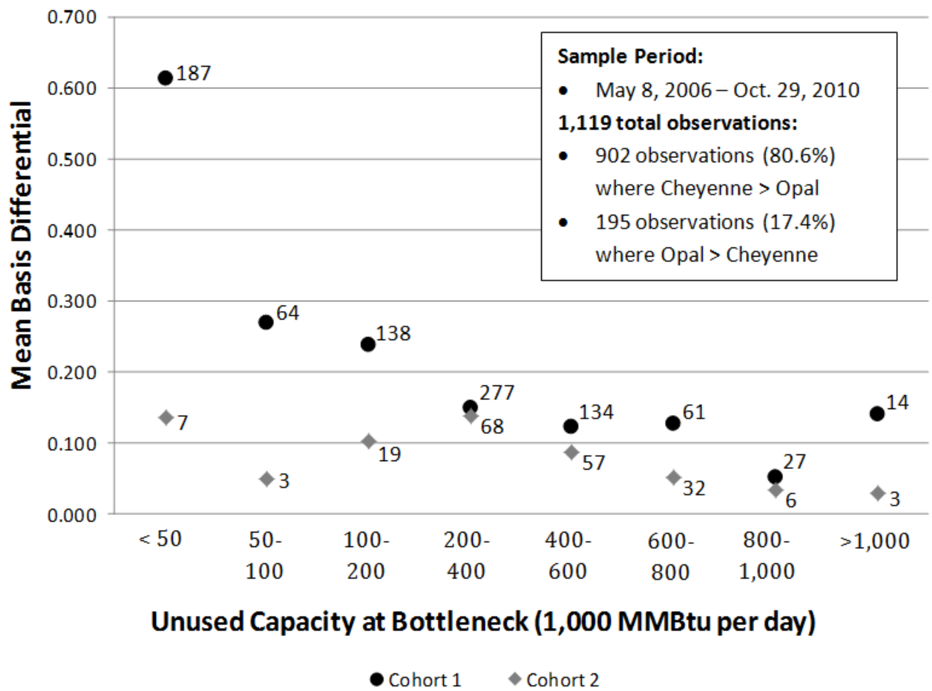

Fig. 3 Relationship between mean Cheyenne-Opal basis differential (absolute value) and unused capacity at the bottleneck (numbers beside each dot represent the number of observations over which that mean is calculated)

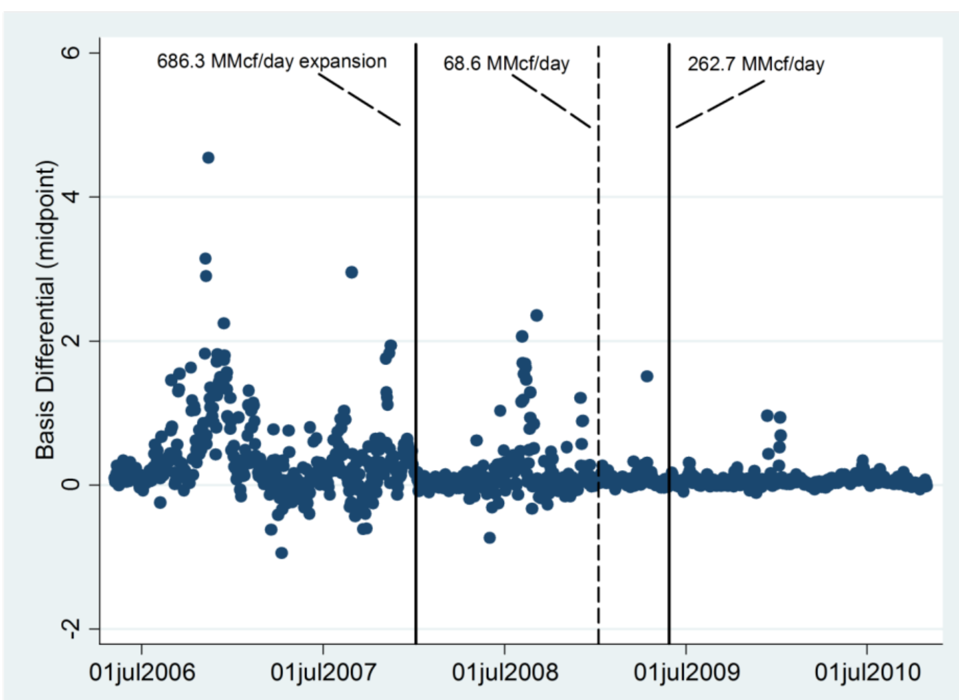

Fig. 4 Cheyenne-Opal basis differential time series with major capacity expansions

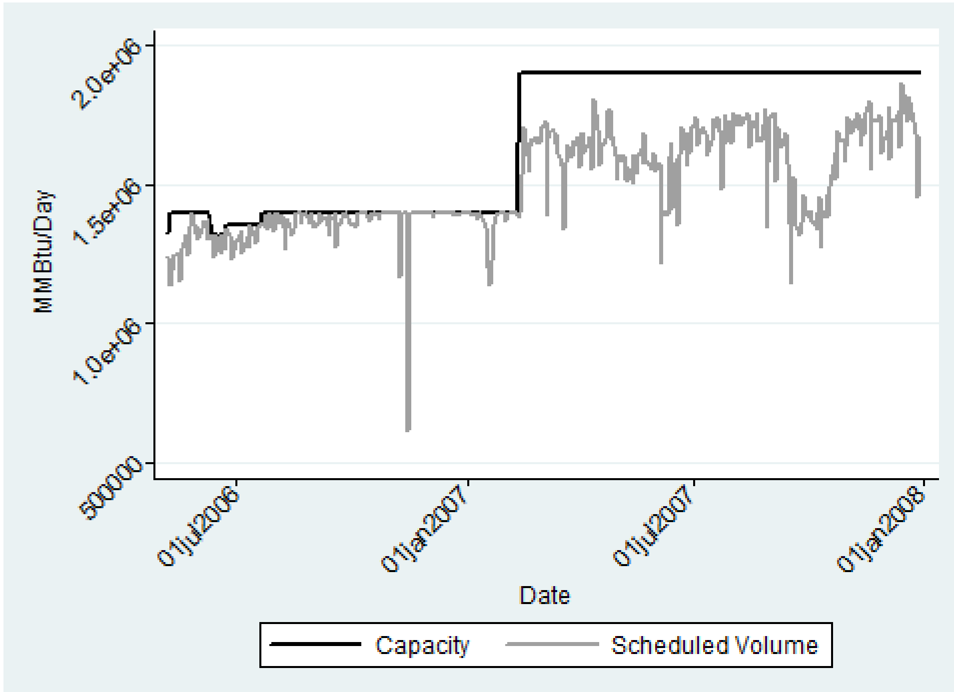

Fig. 5 Capacity expansion event over the bottleneck route

Appendix

Table A.1 2SLS Instrumental Variables Tests (Cohort 1).

Test	**Statistic** (p-value)	**Result**	**Conclusion**
First-Stage Under-Identification Tests			
Angrist-Pischke χ^2 for p^o	124.82 (0.00)	Reject Null	Endogenous regressors sufficiently identified in first stage
Angrist-Pischke χ^2 for p^c	448.73 (0.00)	Reject Null	
Angrist-Pischke χ^2 for τ	109.55 (0.00)	Reject Null	
Angrist-Pischke χ^2 for y^w	429.76 (0.00)	Reject Null	
Angrist-Pischke χ^2 for y^e	511.75 (0.00)	Reject Null	
Angrist-Pischke χ^2 for $stor$	53.94 (0.00)	Reject Null	
First-Stage Weak Identification Tests			
Angrist-Pischke F for p^o	4.38	---	Endogenous regressors strongly identified in first stage, with the possible exceptions of τ and $stor$[1]
Angrist-Pischke F for p^c	15.75	---	
Angrist-Pischke F for τ	3.84	---	
Angrist-Pischke F for y^w	15.08	---	
Angrist-Pischke F for y^e	17.96	---	
Angrist-Pischke F for $stor$	1.89	---	
Second-Stage Weak Identification Tests			
Cragg-Donald Wald F	1.25	---	Inconclusive[2]
Kleibergen-Paap Wald F	1.19	---	Inconclusive[2]
Weak Instrument Robust Inference			
Anderson-Rubin Wald F	2.30 (0.00)	Reject Null	Coefficients of endogenous regressors jointly significant despite potential weak identification
Anderson-Rubin Wald χ^2	77.67 (0.00)	Reject Null	
Stock-Wright LM S statistic	74.22 (0.00)	Reject Null	
Second-Stage Under-Identification Test			
Kleibergen-Paap Rank LM	40.71 (0.04)	Reject Null	Rank condition met
Second-Stage Over-Identification Test			
Hansen's J	33.03 (0.16)	Fail to Reject Null[3]	Instruments valid (jointly uncorrelated with errors)
Exogeneity Test (C-statistic)	1.91 (0.59)	Fail to Reject Null[3]	Production variables sufficiently exogenous
Endogeneity Test (χ^2 statistic)	62.29 (0.00)	Reject Null	Endogenous regressors sufficiently endogenous

[1]First stage Stock-Yogo critical values are not available (Baum et al. 2007), although comparison to the Stock-Yogo critical values for a single endogenous regressor indicates the possibility of weak instrument bias for τ and $stor$ greater than 30% of the bias that would occur using OLS.
[2]The 2SLS Stock-Yogo critical values are unknown for our model (six endogenous regressors and 32 instruments). According to the Stata *ivreg2* help guide, missing values imply that the Stock-Yogo critical values either have not been tabulated or are not applicable. See Baum et al. (2007) for further explanation.
[3]Failure to reject null is the desired outcome.

Table A.2 2SLS Instrumental Variables Tests (Cohort 2).

Test	**Statistic** (p-value)	**Result**	**Conclusion**
First-Stage Under-Identification Tests			
Angrist-Pischke χ^2 for p^o	124.82 (0.00)	Reject Null	Endogenous regressors identified in first stage
Angrist-Pischke χ^2 for p^c	448.73 (0.00)	Reject Null	
Angrist-Pischke χ^2 for τ	109.55 (0.00)	Reject Null	
Angrist-Pischke χ^2 for y^w	429.76 (0.00)	Reject Null	
Angrist-Pischke χ^2 for y^e	511.75 (0.00)	Reject Null	
Angrist-Pischke χ^2 for $stor$	53.94 (0.00)	Reject Null	
First-Stage Weak Identification Tests			
Angrist-Pischke F for p^o	16.67	---	Endogenous regressors strongly identified in first stage, with the possible exceptions of τ and $stor$[1]
Angrist-Pischke F for p^c	10.49	---	
Angrist-Pischke F for τ	1.74	---	
Angrist-Pischke F for y^w	7.17	---	
Angrist-Pischke F for y^e	7.73	---	
Angrist-Pischke F for $stor$	3.00	---	
Second-Stage Weak Identification Tests			
Cragg-Donald Wald F	1.25	---	Inconclusive[2]
Kleibergen-Paap Wald F	1.19	---	Inconclusive[2]
Weak Instrument Robust Inference			
Anderson-Rubin Wald F	2.86 (0.00)	Reject Null	Coefficients of endogenous regressors jointly significant despite potential weak identification
Anderson-Rubin Wald χ^2	120.49 (0.00)	Reject Null	
Stock-Wright LM S statistic	53.37 (0.01)	Reject Null	
Second-Stage Under-Identification Test			
Kleibergen-Paap Rank LM	42.33 (0.03)	Reject Null	Rank condition met
Second-Stage Over-Identification Test			
Hansen's J	26.93 (0.41)	Fail to Reject Null[3]	Instruments valid (jointly uncorrelated with errors)
Exogeneity Test (C-statistic)	5.30 (0.15)	Fail to Reject Null[3]	Production variables sufficiently exogenous
Endogeneity Test (χ^2 statistic)	27.93 (0.00)	Reject Null	Endogenous regressors sufficiently endogenous

[1] See Table A.1, note 1.
[2] See Table A.1, note 2.
[3] See Table A.1, note 3.